

This paper summarizes the changes that are expected to make it into the next version of the C standard (“C23”) from a ISO/IEC TS-18661 relating to the binding of recent versions of IEC 60559 and IEEE 754 into C. In particular:

TS 18661-1: Binary floating-point arithmetic (IEC 60559 requirements)

TS 18661-2: Decimal floating-point arithmetic (IEC 60559 requirements, supersedes ISO/IEC TR 24732)

TS 18661-3: Interchange and extended types (optional by IEC 60559)

TS 18661-4a: Supplementary mathematical functions (optional by IEC 60559, reduction functions under 4b not added to C)

A summary for each TS part is given below (part numbers correspond to the TS name after the “-“).

Part 1: Binary floating point

Macros added to give integer type widths.

Macros and functions added to query and set floating-point environment flags and modes.

Macros and functions added (ex. fromfpx, roundeven, fmaxmag, llogb, nextup, fadd, ffma, totalorder, canonicalize, setpayload, strfromd including tgmth versions).

Constant rounding modes added: #pragma STDC FENV_ROUND direction - Some standard functions are affected by this (Ex. cos, exp, log, scalbn, cbrt, lgamma, rint, fadd, wcstod, wprintf).

Macros added for signaling NaNs.

Macros added for queries on the classification of floating-point values (Ex. iscanonical, issignaling, iszero).

Part 2: Decimal floating point

Distinct types (from float, double and long double) conditionally added for decimal floating-point types: `_Decimal{32,64,128}`.

Literal suffixes conditionally added for decimal floating-point types: `df/DF`, `dd/DD`, `dl/DL`.

Macros conditionally added to provide information about decimal floating-point values (Ex. Min, max values, `DEC_EVAL_METHOD`).

Macros and functions conditionally added to provide decimal floating-point functions and environment modes corresponding to binary floating-point (Ex. `fe_dec_setround`, `DEC_INFINITY`, `cosd32`, `expd128`, `fabsd64`, `lroundd64`, `nextafterd32`, `strtod64`, `dMadddN`, `dMmuldN`).

Functions conditionally added to get decimal floating-point type specific information (Ex. `samequantumd32`, `lquantexpd64`).

Functions conditionally added to convert between different decimal floating-point encodings (Ex. `encodedecd128`, `decodebind64`).

Format specifiers conditionally added to the `printf/scanf` family of functions to handle decimal floating-point types.

Part 3: Interchange (fN, dN) and extended (fNx, dNx) types - conditionally normative annex

Distinct types added for binary and decimal floating-point interchange and extended types (Ex. `_Float32`, `_DecimalN`, `_FloatNx`)

Literal suffixes added for binary floating-point and decimal floating-point types: `fN/FN`, `fNx/FNx`, `dN/DN`, `dNx/DNx`.

Binary and Decimal floating-point information macros generalized to interchange and extended types (Ex. `FLT_N_MAX`, `DEC_NX_TRUE_MIN`).

Binary and Decimal floating-point functions, type generic macros, and other macros generalized to interchange and extended types (Ex. `coshfN`, `ceilfNx`, `sinhdNx`, `dMadddNx`, `strtofN`, `FP_FAST_FMADDFN`, `FLT_N_SNAN`, and `tgmath` versions).

Decimal floating-point specific functions generalized to interchange and extended types (Ex. `encodedecdN`, `quantizedNx`).

Binary complex and imaginary types generalized to interchange and extended types (Ex. `_FloatN_Imaginary`, `_FloatNx_Complex`)

Binary complex floating-point functions generalized to interchange and extended types (Ex. `cexpfN`, `crealfNx`).

Evaluation method macro values updated to include interchange and extended types (`DEC_EVAL_METHOD N` for `_DecimalN`, `FLT_EVAL_METHOD N+1` for `_FloatNx`).

Encoding and decoding functions added to allow conversions with non-arithmetic interchange formats (Ex. `decodefN`, `dMecndecdN`).

General encoding related functions added (Ex. `dMencbindN`, `strfromencdcdN`).

Part 4a: Supplementary math functions

New functions added for standard types and conditionally added for interchange and extended floating-point types (Ex. `pown`, `acospifN`, `exp2m1dN`, `compounddNx`).